

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

Author:

Tahir JAVAID

Supervisor:

Prof. Guoming CHEN

Co-supervisor:

Prof. Mingshui CHEN

*A thesis submitted in fulfillment of the requirements
for the degree of Doctor of Science*

in the

Institute of High Energy Physics
Chinese Academy of Sciences



14 **Contents**

15 **List of Figures** **v**

16 **List of Tables** **vii**

17 **1 Probing four lepton final-state for Higgs Cross section measurement** **1**

18 **1.1 Introduction** **1**

19 **1.1.1 Datasets** **2**

20 List of Figures

21	1.1 Fiducial phase definition among different models.	2
----	---	---

List of Tables

Chapter 1

Probing four lepton final-state for Higgs Cross section measurement

1.1 Introduction

Higgs particle has been the first fundamental scalar particle which was observed at LHC, CERN. Some years ago in 2012, both CMS and ATLAS collaborations announced its discovery [1, 2] with its mass around 125 GeV. Later on, these experiments have also studied its comprehensive properties [3, 4, 5, 6] which are found to be consistent with the SM Higgs.

In this chapter, measurements of fiducial cross sections (inclusive and differential) of Higgs in the $H \rightarrow 4\ell$ ($\ell = e, \mu$) decay channel using full Run 2 data recorded with CMS from collisions of proton-proton at $\sqrt{s} = 13\text{TeV}$ are presented which give a direct test of the perturbative QCD calculations of SM (Higgs sector) at TeV energy scale. Any deviation of the integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. These measurements with earlier or partial Run2 CMS data has also been performed Ref. [Khachatryan:2015yvw], [CMS:2015hja], [CMS:2016rqf], [Sirunyan:2017exp], [CMS:2018hhg]. Nearly similar measurements have already been reported by ATLAS experiment in $H \rightarrow \gamma\gamma$ [atlas-conf-2019-029], [7, 8], $H \rightarrow 4\ell$ [atlas-conf-2019-025], [9, 10] decay channels, combination of both channels [atlas-conf-2019-032], [11, 12] and by CMS experiment in $H \rightarrow \gamma\gamma$ decay channel [13].

The integrated fiducial cross section measurements are performed using collisions data of CMS (LHC) corresponding to 137.1 fb^{-1} integrated luminosity at $\sqrt{s} = 13\text{ TeV}$. It is important to minimize results' dependence on underlying models of Higgs production mechanism and its properties. Hence, cross sections are

48 extracted in the fiducial phase space that closely matches with the detector geom-
 49 etry as shown in Figure 1.1. The remaining model dependence is extracted by
 50 repeating the measurements for SM and other exotic models and reported as a
 51 distinct systematic uncertainty. All measurements includes the corrections for the
 52 finite detection efficiency and resolution effects. As $\sigma \times BR$ strongly depends on
 53 the mass of Higgs, measurements are extracted assuming $m_H = 125.09$ GeV, as
 54 measured from $H \rightarrow 2\gamma$ and $H \rightarrow 4\ell$ decay modes at CMS experiment of LHC
 [14].

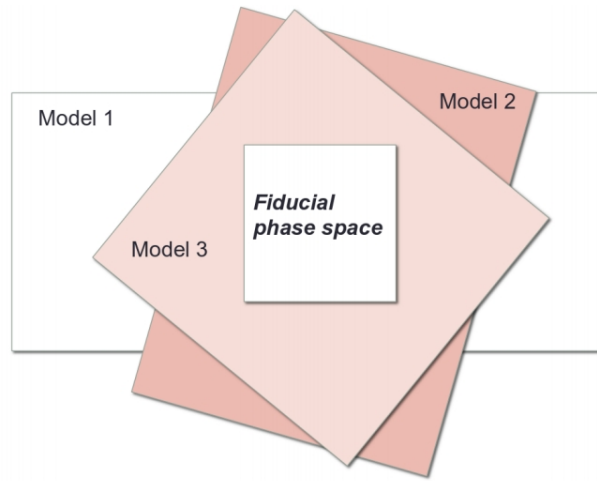


FIGURE 1.1: Fiducial phase definition among different models.

55
 56 The cross sections are also measured differentially as a function of kinematic
 57 observables. In this dissertation we study the differential observables which are
 58 sensitive to production mechanism of Higgs: transverse momentum and the ra-
 59 pidity of Higgs, the transverse momentum of leading jet and jet multiplicity.

60 The structure of this chapter follows as the datasets and MC or simulated sam-
 61 ples used are introduced in Section ???. The analysis strategy is described in Section
 62 ??, where event selection, fiducial phase space definition and strategy method for
 63 extracting inclusive and differential fiducial cross sections. In Section ?? details of
 64 systematic uncertainties affecting the cross section measurement are explained. In
 65 Section ?? results of measurements are compared with SM predictions.

66 1.1.1 Datasets

Bibliography

- [1] ATLAS Collaboration. "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC". In: *Physics Letters B* 716 (2012), pp. 1–29.
- [2] CMS Collaboration. "Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC". In: *Physics Letters B* 716 (2012), pp. 30–61.
- [3] ATLAS Collaboration. "Evidence for the spin-0 nature of the Higgs boson using ATLAS data". In: *Phys. Lett. B* 726 (2013), pp. 120–144.
- [4] ATLAS and CMS Collaboration. "Combined measurement of the Higgs Boson mass in pp collisions at $\sqrt{s} = 7$ and 8 TeV with the ATLAS and CMS experiment". In: *Physics Rev. Lett.* 115 (2015), p. 191803.
- [5] ATLAS Collaboration. "Study of the spin and parity of the Higgs Boson in diboson decays with the ATLAS detector". In: *Eur. Phys. Phys. J. C* 75 (2015), p. 476.
- [6] CMS Collaboration. "Constraints on the spin-parity and anomalous HVV couplings of the Higgs boson in proton collisions at 7 and 8 TeV". In: *Phys. Rev. D* 92 (2015), p. 012004.
- [7] G. Aad et al. "Measurements of fiducial and differential cross sections for Higgs boson production in the diphoton decay channel at $\sqrt{s} = 8$ TeV with ATLAS". In: *JHEP* 2014.9 (2014), pp. 1–61.
- [8] G. Aad et al. "Measurement of the Higgs boson production cross section at 7, 8 and 13 TeV center-of-mass energies in the channel $H \rightarrow \gamma\gamma$ with the ATLAS detector". In: *JHEP* 2014.9 (2014), pp. 1–61.
- [9] G. Aad et al. "Fiducial and differential cross sections of Higgs boson production measured in the four-lepton decay channel in pp collisions at $\sqrt{s} = 8$ TeV with the ATLAS detector". In: *Physics Letters B* 738 (2014), pp. 234–253.

- [10] ATLAS collaboration. “Measurements of the Higgs boson production cross section at 7, 8 and 13 TeV centre-of-mass energies and search for new physics at 13 TeV in the $H \rightarrow ZZ^* \rightarrow \ell^+ \ell^- \ell'^+ \ell'^-$ final state with the ATLAS detector”. In: *ATLAS-CONF-2015-059* (2015).
- [11] G. Aad et al. “Measurements of the Total and Differential Higgs Boson Production Cross Sections Combining the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4\ell$ Decay Channels at $\sqrt{s} = 8$ TeV with the ATLAS Detector”. In: *Phys. Rev. Lett.* 115 (2014), p. 091801.
- [12] ATLAS collaboration. “Measurements of the total cross sections for Higgs boson production combining the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channels at 7, 8 and 13 TeV center-of-mass energies with the ATLAS detector”. In: *ATLAS Note ATLAS-CONF-2015-069* (2015), pp. 1–61.
- [13] V. Khachatryan et al. “Measurement of differential cross sections for Higgs boson production in the diphoton decay channel in pp collisions at $\sqrt{s} = 8$ TeV”. In: *Eur. Phys. J. C* 76 (2016), pp. 1–31.
- [14] CMS Collaboration. “Precise determination of the mass of the Higgs boson and tests of compatibility of its couplings with the standard model predictions using proton collisions at 7 and 8 TeV”. In: *Eur. Phys. J. C* 75 (2015), p. 212.